

IMPROVING NLSAT FOR NONLINEAR REAL ARITHMETIC

Zhonghan Wang

Institute of Software, Chinese Academy of Sciences
University of Chinese Academy of Sciences

wangzhonghan272@gmail.com, zhece.wang@mail.utoronto.ca

Code available: <https://github.com/yogurt-shadow/clauseSMT>

Introduction

The Model-Constructing Satisfiability Calculus (MCSAT) framework has been applied to SMT problems over various arithmetic theories. NLSAT, an implementation using cylindrical algebraic decomposition (CAD) for explanation, is especially competitive for nonlinear real arithmetic (NRA) constraints. However, current Conflict-Driven Clause Learning (CDCL)-style algorithms only consider literal information when making decisions, and thus ignore the influence of clauses on arithmetic variables. This limitation may lead NLSAT to encounter unnecessary conflicts due to suboptimal literal choices. To address this issue, we analyze conflicts caused by literal decisions and incorporate clause-level information that directly affects arithmetic variables. We propose two main algorithmic improvements: a clause-level feasible-set-based look-ahead mechanism and an arithmetic propagation-based branching heuristic. We implement our solver, named clauseSMT, based on a dynamic variable ordering framework. Experiments indicate that clauseSMT is competitive on nonlinear real arithmetic problems compared with existing SMT solvers (CVC5, Z3, YICES2), and it outperforms all of them on satisfiable instances of SMT(QF_NRA) in SMT-LIB. We also evaluate the effectiveness of our proposed methods.

Look-Ahead Mechanism

Example 5.

$$\begin{aligned} c_1 : [-4, -2] \vee [2, 4] &\rightarrow \{[-4, -2] \cup [2, 4]\}, \\ c_2 : [-6, -5] \vee [1, 5] &\rightarrow \{[-6, -5] \cup [1, 5]\} \end{aligned} \bigwedge \rightarrow \{[2, 4]\}$$

This example shows a path case, since the clauses can be satisfied by deciding literals in the green boxes.

Example 6.

$$\begin{aligned} c_1 : [-4, -2] \vee [2, 4] &\rightarrow \{[-4, -2] \cup [2, 4]\}, \\ c_2 : [-6, -5] \vee [5, 6] &\rightarrow \{[-6, -5] \cup [5, 6]\} \end{aligned} \bigwedge \rightarrow \emptyset$$

This example shows a block case: the clauses cannot be satisfied regardless of which literals we decide.

Figure 1: Literal Decision Cases

In our approach, we employ a look-ahead mechanism that first selects a pre-appointed value from the feasible-set. This pre-appointed value is then used to guide the search for a consistent decision path. This ensures that the feasible-sets of all decided literals during the processing procedure intersect at the clause-set level, allowing the arithmetic variable to be assigned the pre-appointed value. In the block case, the processing algorithm behaves identically to NLSAT, eventually triggering the resolve procedure to revise previous arithmetic assignments.

Clause-Level Propagation

Definition

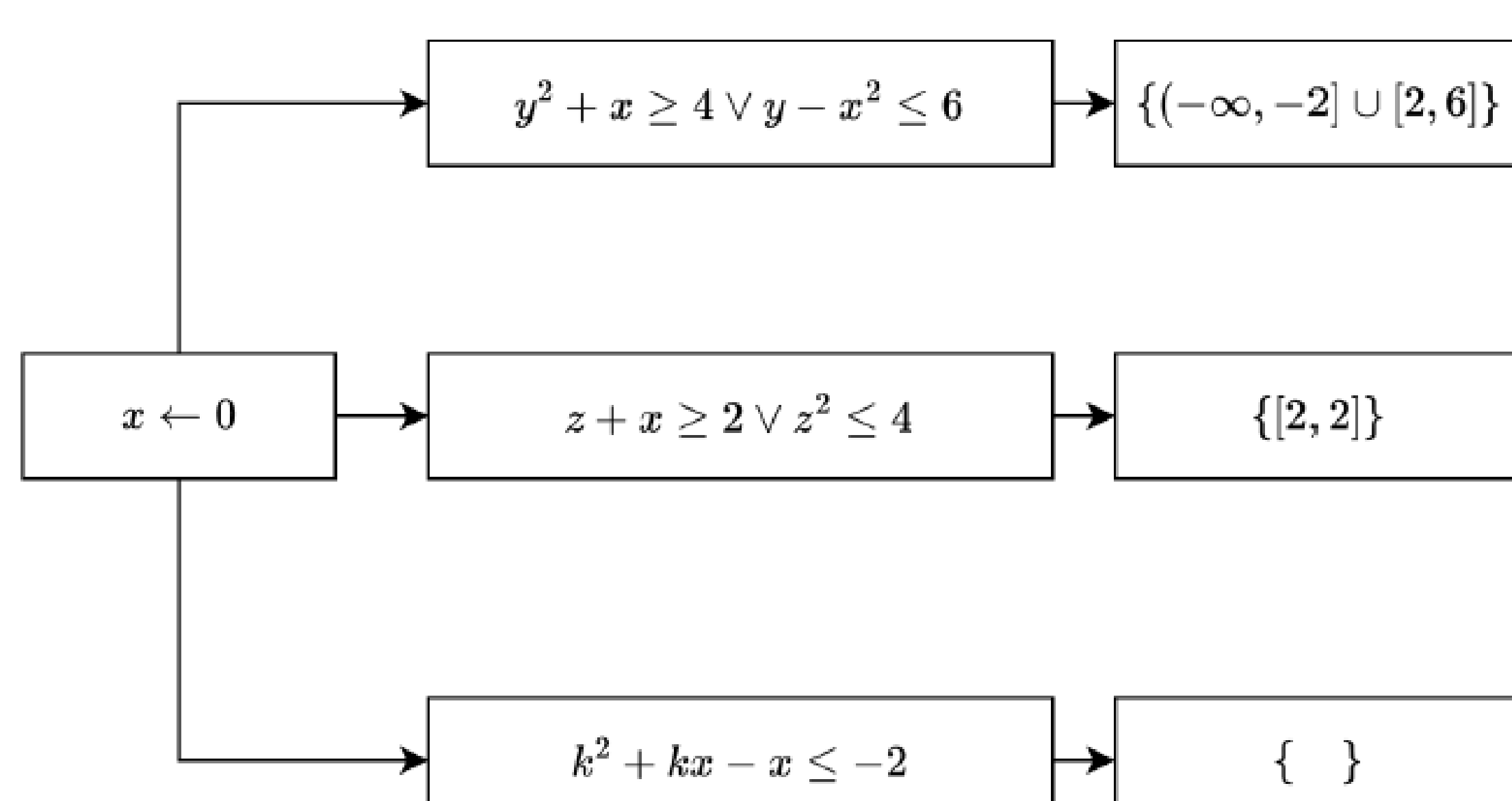


Figure 2: Clause-Level Propagation Cases

Branching Heuristic

We record information about fixed and blocked variables and prioritize them in the branching heuristic, ensuring that the search addresses these critical variables as early as possible

Algorithm 6: Propagation-Based Branching Heuristic

Output: A variable v to branch on

```

1 // Prioritize blocked variables
  (potential conflicts) first
2 if blocked_vars ≠ ∅ then
3   v ← select_from(blocked_vars);
4 // Next, consider fixed variables
  (propagate value)
5 else if fixed_vars ≠ ∅ then
6   v ← select_from(fixed_vars);
  // Otherwise, use VSIDS heuristic for
  normal variables
7 else
8   v ← vsids_select();
9 return v

```

Figure 3: Branching Heuristic

Evaluation

The standard benchmark for evaluating SMT solvers is SMT-LIB.

Category	#inst	Z3	YICES2	CVC5	NLSAT	Ours
20161105-Sturm-MBO	405	SAT	0	0	0	0
		UNSAT	124	285	285	44
		SOLVED	124	285	285	44
20161105-Sturm-MGC	9	SAT	2	0	0	2
		UNSAT	7	0	0	7
		SOLVED	9	0	0	9
20170501-Heizmann	69	SAT	2	0	1	2
		UNSAT	1	12	9	10
		SOLVED	3	12	10	21
20180501-Economics-Mulligan	135	SAT	93	91	89	93
		UNSAT	39	39	35	41
		SOLVED	132	130	124	134
2019-ezsmt	63	SAT	56	52	50	58
		UNSAT	2	2	2	2
		SOLVED	58	54	52	60
20200911-Pine	245	SAT	234	235	199	235
		UNSAT	6	8	5	7
		SOLVED	240	243	204	242
20211101-Geogebra	112	SAT	110	99	91	110
		UNSAT	0	0	0	0
		SOLVED	110	99	91	110
20220314-Uncu	225	SAT	69	70	62	68
		UNSAT	155	153	148	155
		SOLVED	224	223	210	223
hong	20	SAT	0	0	0	0
		UNSAT	8	20	20	12
		SOLVED	8	20	20	12
hycomp	2752	SAT	307	227	225	244
		UNSAT	2242	2201	2212	2088
		SOLVED	2549	2428	2437	2332
kissing	45	SAT	33	10	17	12
		UNSAT	0	0	0	0
		SOLVED	33	10	17	12
LassoRanker	821	SAT	167	122	305	220
		UNSAT	151	260	470	174
		SOLVED	318	382	775	394
meti-tarski	7006	SAT	4391	4369	4343	4391
		UNSAT	2605	2588	2581	2611
		SOLVED	6996	6957	6924	7002
UltimateAutomizer	61	SAT	35	39	35	45
		UNSAT	11	12	10	13
		SOLVED	46	51	45	58
zankl	166	SAT	70	58	58	62
		UNSAT	28	32	32	27
		SOLVED	98	90	90	89
Total	12134	SAT	5569	5372	5475	5541
		UNSAT	5379	5612	5809	5191
		SOLVED	10948	10984	11284	11005

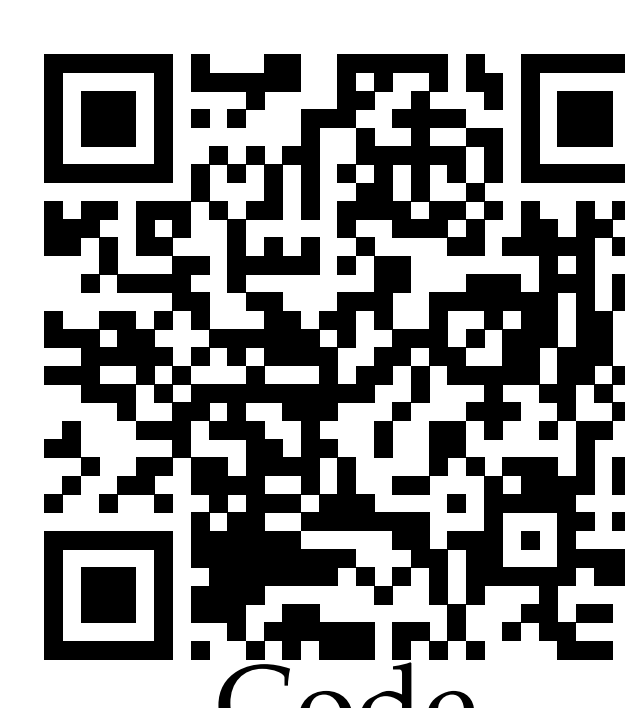
Figure 4: Experiment Result

About Author

The author has been accepted to the PhD program at the University of Toronto but is unable to enroll on time due to security clearance delays. If you are aware of any PhD opportunities or potential collaborations, he would be happy to hear from you.



Paper PDF



Code



author's homepage

I strongly recommend taking a photo.